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Data Analysis Course

Depression Awareness Among Arab Youth:

A Statistical Analysis of Attitudes, Experiences, and Help-Seeking Behaviour

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1. Introduction

The issue of mental well-being is continuously ignored in the realm of public health in the Arab world. Depression is becoming common among adolescents and youth; however, it often goes unnoticed, untreated, and not understood due to cultural stigma, religion-related myths, and insufficient access to professional support services.
The following report provides details of the results obtained through a questionnaire carried out in January 2024 among the Arab youth between the ages of 13 and 19 years. A total of 81 participants responded, and most came from Egypt and Saudi Arabia. This study combines closed questions and open-ended questions to provide insights into the prevalence of depression and how youth perceive and manage their experiences with it.

1.1 Research Hypothesis

The primary hypothesis of this study is:

> *"The majority of Arab teenagers aged 13–19 are personally familiar with symptoms of depression but face significant cultural and social barriers that prevent them from seeking professional help."*

A secondary hypothesis is that respondents significantly underestimate the prevalence of help-seeking behaviour among those with depression, reflecting a broader cultural normalisation of not reaching out.

2. Methodology

2.1 Population of Interest

The target population is Arabic-speaking teenagers and young adults aged 13–19, residing primarily in Egypt, Saudi Arabia, Kuwait, and the wider Gulf and Middle East region. This age group was selected because it represents the transitional life stage during which depression is most likely to go undiagnosed and untreated.

2.2 Sampling Method Snowball Sampling

Snowball sampling was employed as the primary data collection strategy. An initial group of respondents was reached through the researcher's personal and academic networks. Each participant was encouraged to share the Google Form with their own peers, enabling organic growth of the sample.

This method was chosen for two practical reasons:

(1) depression is a sensitive topic, and trust between referrer and respondent increases the likelihood of honest and complete answers;

(2) the researcher's direct network is limited in geographic spread, while snowball propagation allows responses from multiple countries within the same cultural context.

The primary limitation of snowball sampling is that it may over-represent socially connected individuals and under-represent isolated teenagers who may be among the most at-risk. This is acknowledged as a bias source in Section 6.

2.3 Survey Instrument

The survey was built using Google Forms and contained 11 core questions spanning:

- Demographic information: age, country, gender
- Personal experience: sadness frequency, self-harm thoughts, escape thoughts
- Help-seeking behaviour: desire to vent, professional help received
- Beliefs and knowledge: recovery likelihood, % seeking help, highest-risk group
- Opinions: gender differences, primary causes, personal views on depression

Where applicable, Likert-scale responses were numerically encoded to enable descriptive statistics (see Section 3).

2.4 Bias Identification & Mitigation

Several potential biases were identified in the survey design:

- Social desirability bias: Respondents may underreport negative experiences (e.g., self-harm thoughts) to appear socially acceptable. Mitigation: the form was anonymous with no identifying information collected.
- Selection bias: Snowball sampling reaches socially networked individuals. Those who are severely depressed and socially withdrawn may be systematically excluded.
- Gender bias: The sample is 73% female. Female respondents may be more willing to discuss emotional topics, inflating the rates of self-reported depressive experiences relative to the general population.
- Recall bias: Questions about past experiences (e.g., "Have you ever had thoughts of...") rely on memory and may be subject to over- or under-reporting.

2.5 Questions & Survey Link

Demographics:
- Age
- Country of residence
- Gender

Core Questions:
- 1) How often do you feel sad?
- 2) Have you ever had thoughts of hurting yourself or that you would be better off dead?
- 3) Have you ever thought of escaping from home?
- 4) Have you thought about venting to someone about how you feel? (If yes, can you go into further details?)
- 5) Have you got any professional help? (If yes, can you go into further details?)
- 6) How likely do you think it is for someone with depression to fully recover with treatment?
- 7) What percentage of people with depression do you believe seek professional help?
- 8) Which age group do you think is at highest risk for having undiagnosed or untreated depression?
- 9) True or false: Men and women have similar rates of developing depression. (Explain your answer.)
- 10) In your opinion, what is the most factor affecting depression?
- 11) What describes your current opinions around depression?

Online survey link: https://forms.gle/71BwuMUjX8HfXcFq6
Number of samples collected: 81

3. Analysis & Visualizations

3.1 Descriptive Statistics Summary

The table below summarises the key descriptive statistics computed from the collected data. Ordinal Likert responses were encoded on a 1–4 integer scale to enable the calculation of mean, median, and mode.

Variable	Mean / %	Median	Mode
Sadness frequency (encoded 1–4)	2.46	2.0	Sometimes
Recovery likelihood (encoded 1–4)	3.01	3.0	Somewhat likely
Self-harm thoughts	64.2%	—	Yes
Escape thoughts	51.9%	—	Yes
Wanted to vent	69.1%	—	Yes
Got professional help	11.1%	—	No

The mean sadness frequency score of 2.46 (on a scale of 1–4) indicates that the average respondent experiences sadness between "Often" and "Frequently", suggesting a meaningfully elevated baseline emotional burden. The recovery likelihood mean of 3.01 indicates cautious optimism, sitting just above "Somewhat likely."

3.2 Sadness Frequency Distribution

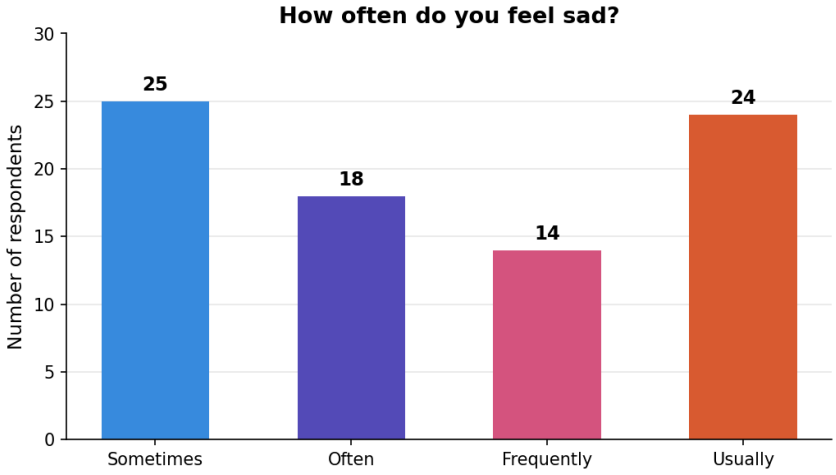

Figure 1 Distribution of sadness frequency among 81 respondents

The distribution of sadness frequency reveals that no single response category dominates. "Sometimes" was the most common answer (25 respondents, 30.9%), followed closely by "Usually" (24, 29.6%), "Often" (18, 22.2%), and "Frequently" (14, 17.3%). The near-equal spread across all four categories is striking: approximately 70% of respondents report feeling sad at a frequency beyond "Sometimes", which is a strong signal that the population is not merely acquainted with occasional sadness but is experiencing sustained emotional distress. The mode of "Sometimes" may reflect an anchoring effect where respondents select the lowest option that still feels true, potentially underreporting their actual experience.

3.3 Yes / No Responses Across Key Questions

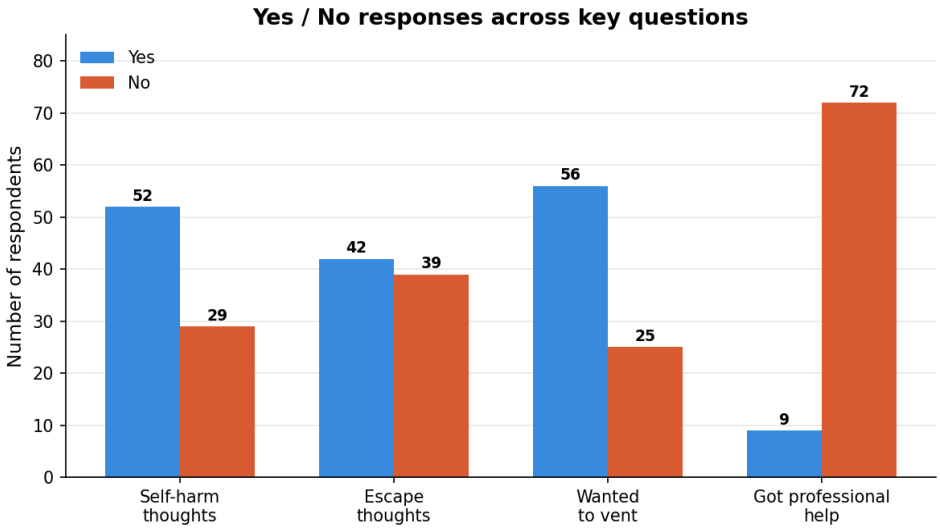

Figure 2 Comparison of Yes/No responses across four key survey questions

Figure 2 highlights a stark contrast between the prevalence of depressive experiences and the rate of professional help-seeking. A significant majority of respondents reported having had thoughts of self-harm (64.2%), thoughts of escaping from home (51.9%), and a desire to vent to someone (69.1%). Against this backdrop, only 9 respondents (11.1%) reported ever receiving professional help. This gap between emotional distress and formal support is one of the most critical findings of the study and directly supports the primary hypothesis. It suggests that the barriers to professional help are not related to a lack of emotional awareness but rather to structural, cultural, and social obstacles.

3.4 Perceived Recovery Likelihood

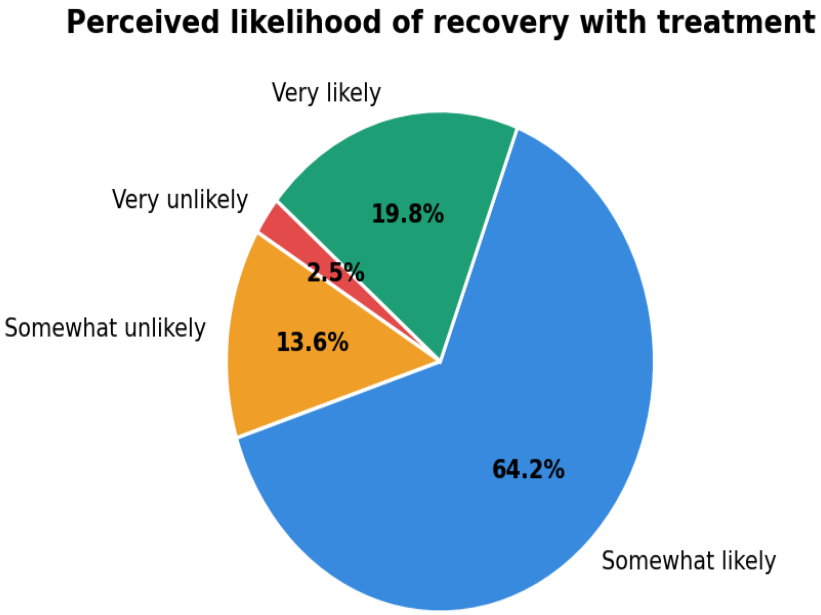

Figure 3 Respondents' beliefs about the likelihood of full recovery with treatment

The majority of respondents (64.2%) selected "Somewhat likely" when asked whether a person with depression could fully recover with treatment. A further 19.8% chose "Very likely", resulting in over 84% of the sample holding an overall optimistic view of treatment outcomes. This finding is encouraging: it suggests that negative attitudes toward treatment efficacy are not a primary barrier to help-seeking. The remaining 13.6% who selected unlikely options may be reflecting personal negative experiences or culturally embedded fatalism. The relatively low standard deviation (0.66) in the encoded recovery score indicates strong consensus around the "Somewhat likely" response.

3.5 Estimated % of Depressed People Who Seek Help

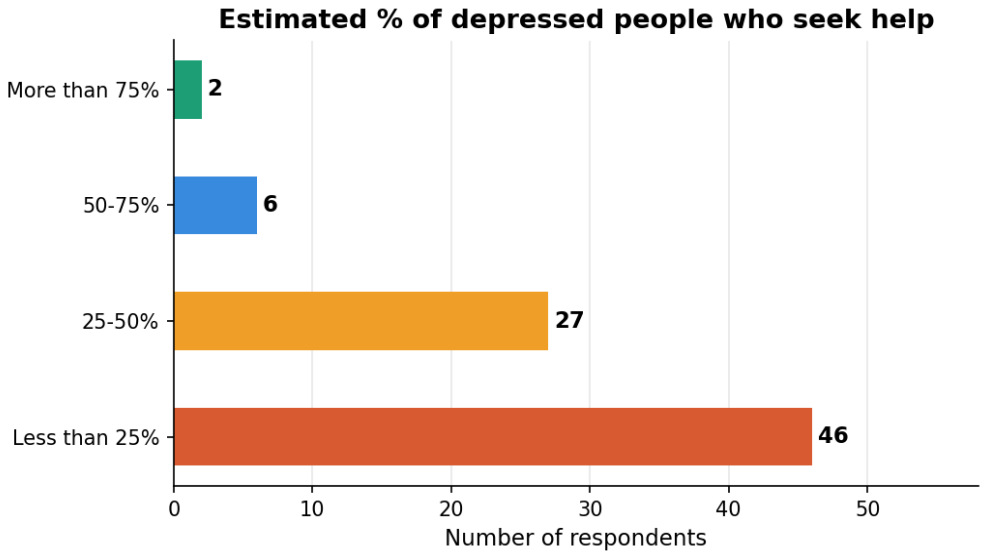

Figure 4 Respondents' beliefs about the proportion of depressed people who seek help

A striking 56.8% of respondents believed that fewer than 25% of people with depression seek professional help. An additional 33.3% estimated the range at 25–50%. In total, approximately 90% of respondents believe that the majority of people with depression do not seek help. This perception is broadly consistent with research literature, which estimates formal treatment rates in the Middle East at below 10% in many populations. The near-unanimous belief that most people do not seek help reflects a cultural awareness of the stigma surrounding mental health a stigma that, as evidenced by the data in Section 3.3, directly affects the respondents themselves.

3.6 Highest-Risk Age Group (Perception)

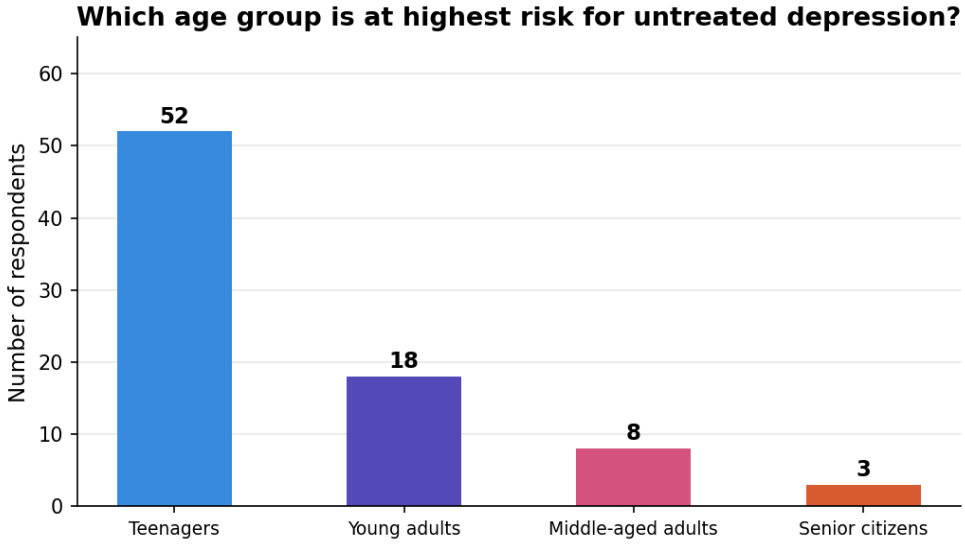

Figure 5 Respondents' perception of the age group most at risk for untreated depression

Teenagers were overwhelmingly identified as the highest-risk group for undiagnosed or untreated depression, with 52 respondents (64.2%) selecting this category. Young adults followed at 22.2%. This result carries a self-referential dimension: the respondents, being predominantly teenagers themselves, may be drawing on their direct social environment. The fact that they correctly identify their own peer group as most vulnerable and yet remain unable to access help underscores the gap between awareness and action discussed throughout this report.

3.7 Gender & Country Distribution

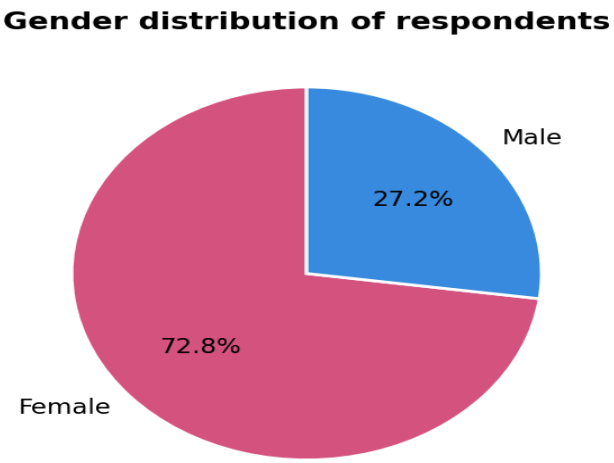

Figure 6 Gender distribution of respondents

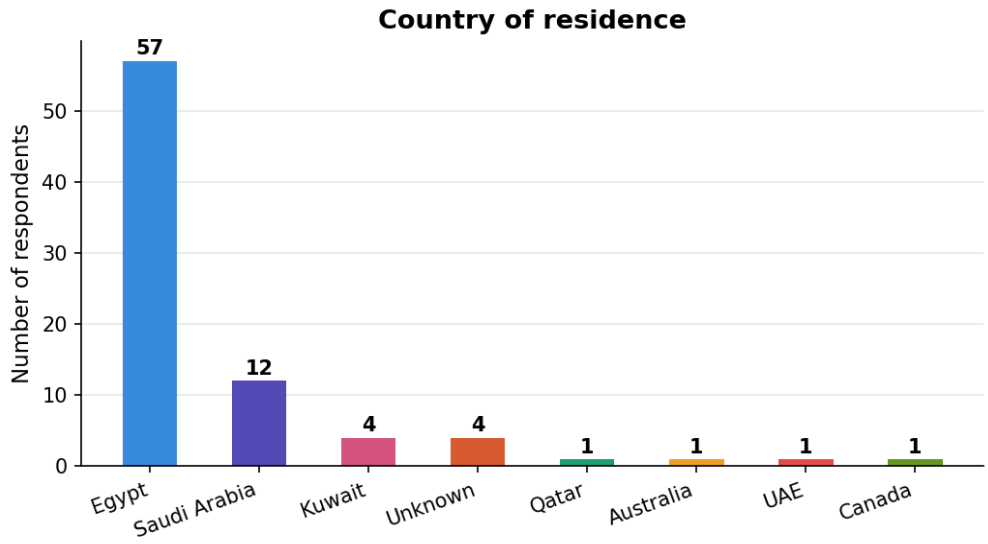

Figure 7 Country of residence of respondents

The sample is 72.8% female (59 respondents) and 27.2% male (22 respondents). This gender imbalance is a known feature of snowball sampling on sensitive topics, as female respondents in the Arab world are generally more willing to engage with emotional and mental health subjects. Geographically, Egypt dominates the sample (57 respondents, 70.4%), followed by Saudi Arabia (12, 14.8%) and Kuwait (4, 4.9%). The concentration in Egypt likely reflects the researcher's personal network and reduces the generalisability of findings to the broader Arab world.

4. Conclusion

The data collected in this survey provides strong support for both hypotheses. The primary hypothesis that Arab teenagers are personally familiar with depressive symptoms yet face significant barriers to professional help is clearly upheld. More than 64% reported self-harm thoughts, more than half considered escaping from home, and nearly 70% wanted to vent to someone. Despite this, fewer than 12% had accessed professional support.

The secondary hypothesis that respondents underestimate how few people seek help is also supported, with 90% of respondents estimating that fewer than half of depressed people seek professional help. This perception reflects genuine awareness of the help-seeking gap, even if respondents are themselves caught within it.

The most consistent qualitative theme across open-ended responses is the barrier posed by family attitudes: many respondents described parents who dismiss mental illness as weakness, laziness, or a spiritual failing. This is not merely a personal opinion it functions as a structural barrier, preventing teenagers from disclosing their experiences and, consequently, from accessing care.

Importantly, the data does not support pessimism about treatment. The majority of respondents believe recovery is possible. The problem is not a lack of belief in treatment it is a lack of access, trust, and social permission to seek it.

5. Critical Reflection

Several limitations and hidden biases emerged upon reviewing the collected data:

- Gender skew: The 73% female composition means the sample is unlikely to capture the male experience of depression, which qualitative responses suggest is characterised by greater concealment and social pressure to appear strong. A targeted sub-survey of male respondents would be a valuable extension.
- Geographic concentration: 70% of respondents are from Egypt. While Egypt is culturally representative of parts of the Arab world, the findings cannot be generalised to Gulf countries, the Levant, or North Africa without caution.
- Self-selection: Respondents who chose to complete a detailed survey on depression may be more emotionally aware and/or more personally affected than the broader teenage population. This could inflate prevalence estimates.
- Ordinal encoding assumption: Converting Likert responses to integers (e.g., Sometimes=1, Usually=4) assumes equal spacing between categories, which may not hold. This is a standard trade-off in survey research and is acknowledged as an approximation.
- Open-text richness vs. representativeness: Several respondents provided extraordinarily detailed personal accounts spanning multiple paragraphs. While these are invaluable for qualitative insight, they are not statistically representative and should be interpreted as illustrative rather than generalisable.

Despite these limitations, the survey achieves its core objective: generating real, self-reported data that reveals the lived experience of depression among Arab youth, and quantifying the gap between emotional distress and formal help-seeking. Future research should seek larger and more gender-balanced samples, and consider partnering with schools or community organisations to reach those least likely to participate in snowball surveys.

End of Report